

Shock Transmission to USD/VND Exchange Rate Volatility: Evidence from EGARCH-X and SVAR Models

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Abstract—This study examines shock transmission to USD/VND exchange rate volatility using daily data from 2010 to 2026. USD/VND log returns are constructed to capture short-run exchange rate movements, while changes in the VIX, U.S. 10-year Treasury yield, the U.S. Dollar Index, oil prices, and Vietnamese equity-market returns are used to represent global and domestic shock channels. The empirical strategy combines preliminary time-series diagnostics, GARCH-family volatility models, EGARCH-X specifications, HAC regressions, and weekly SVAR analysis. The results show that USD/VND returns exhibit volatility clustering, fat-tailed behavior, and conditional heteroskedasticity. Among the competing models, the EGARCH-based specification provides the most suitable framework for capturing persistent and asymmetric exchange rate volatility. The extended evidence suggests that global financial shocks play an important role in explaining USD/VND volatility, while domestic market conditions provide additional but more limited explanatory power.

Index Terms—Time Series, USD/VND; exchange rate volatility; GARCH-type models; global shocks; Vietnam; financial risk.

I. INTRODUCTION

Exchange rate volatility is an important issue for developing economies because exchange rate fluctuations can affect inflation, trade competitiveness, capital flows, and macroeconomic stability. In countries with less developed financial markets, external financial shocks may be transmitted more strongly due to limited hedging instruments and weaker financial buffers.

To examine exchange rate volatility in developing economies, this study follows the theoretical insight of Aghion et al. (2009), who suggest the following: “The volatility of real shocks relative to financial shocks, which features so prominently in the literature on developed country exchange rate regimes, also matters for developing countries. But because financial shocks tend to be greatly amplified in financially underdeveloped economies, one has to adjust calibrations accordingly.” Based on this argument, the study separates the role of global financial conditions from domestic macro-financial factors in explaining exchange rate volatility. The global variables include the implied volatility index (VIX), used as a proxy for global risk aversion, and the US Dollar Index (DXY), which captures broad movements in the value of the US dollar. The domestic variables include short-term interest rates and inflation expectations, reflecting internal monetary conditions and macroeconomic expectations.

Empirically, this research applies a GARCH model to daily exchange rate data from August, 2000 to January, 2026. The GARCH framework is appropriate because exchange rate returns often exhibit volatility clustering and time-varying conditional variance. By incorporating both global and domestic

variables, the study aims to evaluate whether external financial shocks or domestic macro-financial conditions play a more important role in explaining exchange rate volatility.

II. LITERATURE REVIEW

Exchange rate volatility has been widely examined in international finance because it affects trade, investment, inflation, and financial stability, especially in developing economies. Aghion et al. (2009) argue that the impact of exchange rate volatility depends on the level of financial development. In countries with less developed financial systems, exchange rate fluctuations may have stronger negative effects because firms have limited access to hedging instruments and external finance. This implies that developing economies may be more vulnerable to both real and financial shocks than advanced economies.

From a methodological perspective, exchange rate volatility is commonly modeled using ARCH and GARCH-type models. Engle (1982) introduced the ARCH model to capture time-varying conditional variance, while Bollerslev (1986) extended this framework into the GARCH model by allowing current volatility to depend on both past shocks and past volatility. These models are appropriate for exchange rate data because exchange rate returns often show volatility clustering, fat tails, and conditional heteroskedasticity.

Recent literature has moved beyond simply estimating volatility and has focused on identifying its causes. Ozkaya and Altun (2024), in *Domestic and Global Causes for Exchange Rate Volatility: Evidence From Turkey*, examine the exchange rate volatility of the Turkish lira by separating domestic and global financial determinants. Their study applies a GARCH(1,1) model and Lyapunov exponents over the period from March 1, 2019 to November 11, 2021. The authors include global variables such as the VIX and dollar index, and domestic variables such as CDS and inflation expectation differentials. Their GARCH results show that Turkish exchange rate volatility is significantly affected by VIX and CDS, indicating that both global risk aversion and domestic financial risk contribute to currency instability.

Other studies also emphasize the importance of external financial conditions in explaining exchange rate volatility. Caporale et al. (2017) examine the effect of equity and bond portfolio inflows on exchange rate volatility in seven Asian developing and emerging economies, including India, Indonesia, Pakistan, the Philippines, South Korea, Taiwan, and Thailand. Using GARCH models and Markov-switching specifications, their study shows that capital flows are an important source of

exchange rate volatility in emerging markets. This supports the view that exchange rates in developing economies are sensitive to global liquidity and investor sentiment.

For Vietnam, previous research has mostly studied the consequences of exchange rate volatility rather than its determinants. Thuy and Thuy (2019) examine the impact of exchange rate volatility on Vietnam's exports using quarterly data from 2000 to 2014 and an ARDL approach. Their study focuses on how volatility affects trade performance rather than how domestic and global variables generate exchange rate fluctuations. Similarly, Huynh (2019) analyzes the effect of exchange rate volatility on Vietnam's bilateral imports and uses GARCH as one measure to construct exchange rate volatility. These studies confirm the relevance of exchange rate volatility for Vietnam's external sector, but they do not directly investigate the global and domestic sources of exchange rate volatility.

Overall, the literature suggests three main points. First, exchange rate volatility is particularly important for developing economies because financial underdevelopment may amplify the impact of shocks. Second, GARCH-type models provide a suitable framework for modeling time-varying exchange rate volatility. Third, recent studies highlight the need to distinguish between global and domestic causes of volatility. Following Ozkaya and Altun (2024), this study examines whether global financial variables, such as VIX and DXY, and domestic financial variables, such as short-term interest rates and inflation expectations, explain exchange rate volatility in Vietnam. By applying a GARCH(1,1) framework to the Vietnamese exchange rate market, this study contributes to the literature by extending the domestic–global volatility framework to a developing economy with a managed floating exchange rate regime.

III. DATA AND METHODOLOGY

A. Data and Variable Construction

This study uses daily time-series data from 2010 to 2026 to examine the transmission of global and domestic financial shocks to USD/VND exchange rate volatility. The main variable of interest is the USD/VND nominal exchange rate, which represents the value of the Vietnamese dong relative to the U.S. dollar. The explanatory variables include the CBOE Volatility Index (VIX), the U.S. 10-year Treasury yield, the U.S. Dollar Index (DXY), WTI crude oil prices, and the Vietnam stock market index (VN-Index). These variables are chosen because they represent the main external and domestic channels through which financial shocks may affect Vietnam's foreign exchange market.

The VIX is used as a proxy for global risk aversion and international financial uncertainty. In periods of heightened uncertainty, capital flows and exchange rate expectations may become more volatile, increasing pressure on emerging-market currencies. The U.S. 10-year Treasury yield captures changes in global interest-rate conditions, which can influence international capital allocation and the relative attractiveness of U.S. dollar assets. The DXY measures broad movements in the value of the U.S. dollar against major currencies and is therefore directly

relevant to USD/VND exchange rate movements. WTI crude oil prices capture commodity-market shocks, which may affect trade balances, inflation expectations, and macroeconomic conditions. Finally, the VN-Index represents domestic financial-market conditions and is included to distinguish local market movements from global financial shocks.

Because exchange rate levels are usually non-stationary and display strong persistence, the analysis is conducted using return and change variables rather than raw price levels. Modelling returns is also more appropriate for volatility analysis because volatility refers to the conditional variability of short-run changes rather than the level of the exchange rate itself. The daily USD/VND return is defined as

$$r_t = 100 [\ln(USD/VND_t) - \ln(USD/VND_{t-1})]. \quad (1)$$

Price-based variables are transformed into percentage log returns, while index-level and interest-rate variables are transformed into first differences:

$$\begin{aligned} dxy_return_t &= 100\Delta \ln(DXY_t), \\ oil_return_t &= 100\Delta \ln(OIL_t), \\ vnindex_return_t &= 100\Delta \ln(VNINDEX_t), \\ vix_change_t &= \Delta VIX_t, \\ us10y_change_t &= \Delta US10Y_t. \end{aligned} \quad (2)$$

This transformation strategy follows the nature of each variable. Exchange rates, stock indices, oil prices, and the dollar index are price-level variables, so log returns are used to measure proportional changes. In contrast, VIX and U.S. Treasury yields are already expressed as index or rate variables, so first differences are used to capture daily shocks.

To provide preliminary measures of exchange rate variability before formal GARCH estimation, absolute returns, squared returns, and 22-day rolling volatility are also constructed:

$$\begin{aligned} a_t &= |r_t|, \\ q_t &= r_t^2, \\ \sigma_t^{(22)} &= \left[\frac{1}{21} \sum_{j=0}^{21} \left(r_{t-j} - \bar{r}_t^{(22)} \right)^2 \right]^{1/2}. \end{aligned} \quad (3)$$

where a_t , q_t , and $\sigma_t^{(22)}$ denote the absolute return, squared return, and 22-day rolling volatility, respectively. These variables are not treated as substitutes for conditional volatility models. Instead, they are used for descriptive analysis and preliminary diagnostic tests, especially for detecting volatility clustering in the exchange rate return series.

Event dummy variables are included to control for major shock periods relevant to the 2010–2026 sample, including the 2011 Vietnam devaluation episode, the COVID-19 period, and the 2022–2023 Federal Reserve tightening cycle. These dummy variables are used only when they are relevant to the corresponding model specification. Their purpose is not to replace financial shock variables, but to control for large

discrete policy or crisis episodes that may otherwise distort the estimated volatility process.

TABLE I
MAIN VARIABLES USED IN THE STUDY

Variable	Transformation	Interpretation
fx_return	$100\Delta \ln(USD/VND_t)$	Daily USD/VND return.
vix_change	ΔVIX_t	Global risk-aversion shock.
us10y_change	$\Delta US10Y_t$	U.S. interest-rate shock.
dxy_return	$100\Delta \ln(DXY_t)$	U.S. dollar strength shock.
oil_return	$100\Delta \ln(OIL_t)$	Global oil-price shock.
vnindex_return	$100\Delta \ln(VNINDEX_t)$	Domestic stock-market shock.
abs_return	$ fx_return_t $	Absolute return volatility proxy.
squared_return	$(fx_return_t)^2$	Squared return volatility proxy.
rolling_vol_22d	$sd_{22}(fx_return_t)$	22-day rolling volatility.

B. Preliminary Time-Series Diagnostics

Before estimating the volatility models, the transformed variables are examined using descriptive statistics, graphical inspection, and standard time-series diagnostic tests. This step is necessary because GARCH-family models are appropriate only when the return series displays empirical features such as stationarity, volatility clustering, conditional heteroskedasticity, and non-normality. The diagnostics therefore establish whether the modelling strategy is statistically justified before the main models are estimated.

Stationarity is examined using the Augmented Dickey–Fuller (ADF) test and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test. The ADF test evaluates the null hypothesis that a series contains a unit root, while the KPSS test evaluates the null hypothesis that a series is stationary. Using both tests is useful because they approach the stationarity problem from opposite null hypotheses. If the transformed variables reject the ADF unit-root null and do not strongly reject the KPSS stationarity null, they can be treated as suitable inputs for return-based time-series modelling.

Serial correlation is examined using the autocorrelation function (ACF), partial autocorrelation function (PACF), and the Ljung–Box test. ACF and PACF plots provide a visual inspection of short-run dependence in USD/VND returns, while the Ljung–Box test formally evaluates whether autocorrelations are jointly different from zero. The Ljung–Box statistic is defined as

$$Q(m) = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}, \quad (4)$$

where T is the sample size, m is the selected lag length, and $\hat{\rho}_k$ is the sample autocorrelation at lag k . The test is first applied to USD/VND returns to examine serial dependence in the conditional mean. It is then applied to absolute and squared returns to detect volatility clustering. Significant autocorrelation in $|r_t|$ or r_t^2 indicates that large exchange rate movements tend to be followed by large movements, which is a key motivation for conditional volatility modelling.

Conditional heteroskedasticity is examined using the ARCH-LM test. The auxiliary regression is

$$\hat{\varepsilon}_t^2 = a_0 + \sum_{i=1}^q a_i \hat{\varepsilon}_{t-i}^2 + e_t, \quad (5)$$

with the test statistic $LM = TR^2$. Significant ARCH effects indicate that the variance of USD/VND returns is time-varying and depends on past shocks. This result would make constant-variance models inappropriate and support the use of ARCH/GARCH-type specifications.

The Jarque–Bera test is used to examine whether USD/VND returns follow a normal distribution. Financial return series often display excess kurtosis and fat tails, meaning that extreme observations occur more frequently than under the normal distribution. If normality is rejected, Student- t innovations are preferred because they allow the model to accommodate heavier-tailed return behavior. Correlation and variance inflation factor (VIF) checks are also conducted to assess multicollinearity among the shock variables. This is important because highly collinear regressors may make it difficult to separate the volatility effect of individual shocks.

TABLE II
PRE-ESTIMATION DIAGNOSTIC TESTS

Test	Purpose
ADF and KPSS	Test stationarity of transformed variables.
ACF and PACF	Inspect short-run dependence in USD/VND returns.
Ljung–Box test	Test serial correlation in returns.
Ljung–Box on $ r_t $ and r_t^2	Detect volatility clustering.
ARCH-LM test	Test conditional heteroskedasticity.
Jarque–Bera test	Test non-normality and fat-tailed behavior.
Correlation and VIF	Check multicollinearity among shock variables.

C. Conditional Mean Specification

The volatility models are estimated after accounting for short-run dependence in USD/VND returns. This is necessary because volatility models should be applied to innovations after predictable mean dynamics have been filtered out. If serial correlation remains in the return equation, the estimated conditional variance may partly reflect misspecified mean dynamics rather than true volatility persistence.

The conditional mean equation follows an ARMA structure:

$$r_t = \mu + \sum_{i=1}^p \phi_i r_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t, \quad (6)$$

where r_t is the USD/VND return, μ is the constant term, p is the autoregressive order, q is the moving-average order, and ε_t is the innovation. The ARMA order is selected using information criteria and residual diagnostic tests. In the implementation, the ARMA model is used as a mean-filtering step: the selected ARMA residuals are then passed into the GARCH-family models with a zero conditional mean. This keeps the interpretation clear: ARMA captures short-run predictability in returns, while GARCH-type models capture time-varying volatility in the filtered innovations.

The innovation is decomposed as

$$\varepsilon_t = z_t \sqrt{h_t}, \quad z_t \sim t_\nu, \quad (7)$$

where h_t is the conditional variance and z_t follows a Student- t distribution with ν degrees of freedom. The Student- t assumption is used because the diagnostic tests and descriptive statistics of financial returns commonly show fat-tailed behavior. Under this specification, h_t becomes the object of interest because it represents the conditional volatility of USD/VND returns.

D. GARCH-family Volatility Models

To model time-varying exchange rate volatility, this study compares standard GARCH, EGARCH, and GJR-GARCH specifications. These models are appropriate for financial return data because they allow the conditional variance to evolve over time as a function of past shocks and past volatility. The standard GARCH model captures volatility clustering and persistence, while EGARCH and GJR-GARCH allow for asymmetric volatility responses.

The GARCH(1,1) model is written as

$$h_t = \omega + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1}, \quad (8)$$

where α captures the short-run effect of past shocks and β measures volatility persistence. The ARCH term ε_{t-1}^2 measures the effect of recent unexpected movements in exchange rate returns, while the GARCH term h_{t-1} captures the persistence of previous volatility. A high value of $\alpha + \beta$ implies that volatility shocks decay slowly, meaning that the USD/VND market remains volatile for an extended period after large shocks.

The EGARCH(1,1) model is used to capture asymmetric volatility responses:

$$\ln h_t = \omega + \beta \ln h_{t-1} + \alpha (|z_{t-1}| - E|z_{t-1}|) + \gamma z_{t-1}. \quad (9)$$

The coefficient γ captures asymmetry. A statistically significant γ indicates that positive and negative exchange rate shocks affect volatility differently. This is important in the USD/VND context because depreciation and appreciation shocks may not have symmetric effects on market expectations, policy responses, and hedging behavior. Because the EGARCH model is expressed in logarithmic variance, it does not require the same non-negativity restrictions as the standard GARCH model.

The GJR-GARCH(1,1) model is also estimated as an alternative asymmetric specification:

$$h_t = \omega + \alpha \varepsilon_{t-1}^2 + \gamma I_{t-1} \varepsilon_{t-1}^2 + \beta h_{t-1}, \quad (10)$$

where $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$, and $I_{t-1} = 0$ otherwise. The coefficient γ measures the additional volatility effect of negative shocks. Although the conventional leverage interpretation is more common in equity markets, the model is still useful here because it tests whether exchange rate innovations of different signs have unequal effects on conditional volatility.

The competing models are evaluated using log-likelihood, AIC, BIC, convergence status, and post-estimation residual diagnostics. The information criteria are defined as

$$AIC = -2\ell + 2k, \quad BIC = -2\ell + k \ln(T), \quad (11)$$

where ℓ is the maximized log-likelihood, k is the number of estimated parameters, and T is the sample size. Lower AIC and BIC values indicate a better balance between model fit and model complexity. However, model selection is not based on information criteria alone. A model is considered suitable only if it converges properly and removes most remaining autocorrelation and ARCH effects from the standardized residuals. Models that fail convergence or produce unstable residual diagnostics are not interpreted, even if their information criteria appear favorable.

E. EGARCH-X Shock Transmission Model

After selecting the preferred baseline volatility specification, the model is extended into an EGARCH-X framework to examine whether global and domestic shocks help explain USD/VND volatility. This is the main shock-transmission model of the study. Unlike the baseline GARCH-family models, which explain volatility only through past innovations and past volatility, the EGARCH-X model allows external and domestic financial variables to enter the conditional variance equation directly.

The shock variables are standardized before entering the variance equation:

$$X_t = \begin{bmatrix} Z_t^{VIX} \\ Z_t^{US10Y} \\ Z_t^{DXY} \\ Z_t^{OIL} \\ Z_t^{VNI} \end{bmatrix}. \quad (12)$$

Standardization makes the shock coefficients more comparable because the variables are measured in different units. For example, changes in VIX, changes in U.S. yields, and log returns on DXY or VN-Index do not share the same natural scale. After standardization, the estimated coefficients can be interpreted as the volatility effect associated with a one-standard-deviation movement in the corresponding shock variable.

The EGARCH-X model is specified as

$$\begin{aligned} \ln h_t = & \omega + \beta \ln h_{t-1} + \alpha (|z_{t-1}| - E|z_{t-1}|) + \gamma z_{t-1} \\ & + \delta_1 Z_t^{VIX} + \delta_2 Z_t^{US10Y} + \delta_3 Z_t^{DXY} \\ & + \delta_4 Z_t^{OIL} + \delta_5 Z_t^{VNI} + \sum_{m=1}^M \lambda_m D_{m,t}. \end{aligned} \quad (13)$$

Here, $D_{m,t}$ denotes event dummy variables included in the selected specification. The coefficients $\delta_1, \dots, \delta_5$ measure the contribution of global and domestic financial shocks to exchange rate volatility. A positive and statistically significant coefficient implies that the corresponding shock increases

conditional volatility, while an insignificant coefficient indicates weak evidence of a direct volatility effect in the selected specification.

The EGARCH-X specifications are estimated in several forms to distinguish the direction and magnitude of shocks. Signed shock variables are used to test whether the direction of a shock matters, while absolute shock variables are used to test whether the size of the shock matters regardless of direction. Lagged shock variables are included because the USD/VND market may not fully absorb global and domestic financial shocks on the same trading day. This is especially relevant for a managed exchange rate environment, where adjustment may occur gradually rather than instantaneously.

After estimation, the fitted conditional volatility is extracted as

$$\hat{\sigma}_t = \sqrt{\hat{h}_t}. \quad (14)$$

The logarithm of the selected conditional volatility is then used in the supplementary shock-transmission analysis:

$$\log_selected_cond_vol_t = \ln(\hat{\sigma}_t). \quad (15)$$

Taking the logarithm of conditional volatility improves scale stability and makes the series more suitable for subsequent regression and VAR/SVAR analysis. The extracted volatility series represents the model-implied time-varying exchange rate risk after accounting for volatility persistence, fat-tailed innovations, and asymmetric responses.

F. HAC Shock-Volatility Regression

As a supplementary check, this study estimates a shock-volatility regression using heteroskedasticity- and autocorrelation-consistent standard errors. The dependent variable is the selected conditional volatility from the preferred EGARCH-type model. This regression is not designed to replace the EGARCH-X model. Instead, it provides an additional reduced-form check on whether the same shock variables are associated with the estimated volatility series when robust standard errors are used.

The regression is written as

$$\begin{aligned} \log(\hat{\sigma}_t^{sel}) = & c + \beta_1 \Delta VIX_t + \beta_2 \Delta US10Y_t \\ & + \beta_3 r_t^{DXY} + \beta_4 r_t^{OIL} + \beta_5 r_t^{VNI} \\ & + \sum_{m=1}^M \eta_m D_{m,t} + u_t. \end{aligned} \quad (16)$$

HAC standard errors are used because the dependent variable is a generated volatility series and may still exhibit serial dependence and heteroskedasticity. Ordinary least squares standard errors would be too restrictive in this context. Therefore, the HAC specification provides a more conservative inference framework for evaluating whether global and domestic shocks are statistically associated with exchange rate volatility. In the implementation, both signed and absolute lagged shock specifications are considered, with crisis and devaluation controls included when relevant.

The results from this regression are interpreted as supporting evidence rather than structural causal estimates. If the HAC regression produces signs and significance patterns consistent with the EGARCH-X model, it strengthens the interpretation that the identified shocks are related to USD/VND volatility. If the results differ, the EGARCH-X model remains the preferred specification because it models volatility directly within the conditional variance equation.

G. Weekly VAR, Granger Causality, and SVAR Analysis

The final part of the methodology uses weekly VAR and SVAR models as supplementary evidence on dynamic shock transmission. The purpose of this step is to examine whether global and domestic shock variables help predict USD/VND returns and selected exchange rate volatility over time, and how the exchange rate market responds dynamically to identified structural shocks.

Daily variables are aggregated to weekly frequency to reduce high-frequency noise and improve the stability of the system. This aggregation is appropriate because daily exchange rate movements in a managed or partially managed foreign exchange market may be affected by short-run market frictions, policy smoothing, and non-trading-day effects. Weekly aggregation reduces these distortions while preserving the short-run dynamics needed for shock-transmission analysis.

Two VAR systems are estimated: one with USD/VND returns as the target variable and one with selected conditional volatility as the target variable. For the exchange rate return system, the weekly vector is

$$Y_t^{FX} = \begin{bmatrix} vix_change_t \\ us10y_change_t \\ dxy_return_t \\ oil_return_t \\ vnindex_return_t \\ fx_return_t \end{bmatrix}. \quad (17)$$

For the volatility system, the weekly vector is

$$Y_t^{VOL} = \begin{bmatrix} vix_change_t \\ us10y_change_t \\ dxy_return_t \\ oil_return_t \\ vnindex_return_t \\ log_selected_cond_vol_t \end{bmatrix}. \quad (18)$$

The reduced-form VAR model is

$$Y_t = c + \sum_{i=1}^p A_i Y_{t-i} + u_t, \quad (19)$$

where Y_t is the vector of weekly variables, c is the vector of constants, A_i is the coefficient matrix at lag i , and u_t is the vector of reduced-form residuals. The lag length is selected using information criteria and diagnostic checking. In the empirical implementation, the maximum lag length is restricted to keep the weekly system parsimonious and to avoid overfitting relative to the available sample size.

Granger causality tests are applied before the SVAR interpretation to examine whether past values of the shock variables help predict USD/VND returns or selected exchange rate volatility. These tests are interpreted as predictive evidence rather than proof of structural causality. A statistically significant Granger causality result means that past values of a shock variable contain information useful for predicting the target variable, conditional on the other variables in the VAR system.

The structural form is written as

$$Au_t = B\varepsilon_t, \quad (20)$$

where u_t is the vector of reduced-form residuals and ε_t is the vector of structural shocks. The identification ordering places global financial variables before domestic and exchange-rate variables:

$$\begin{aligned} VIX &\rightarrow US10Y \rightarrow DXY \rightarrow OIL \\ &\rightarrow VNINDEX \rightarrow FX/VOL. \end{aligned} \quad (21)$$

This ordering assumes that global shocks are more exogenous to Vietnam's exchange rate market than domestic financial variables. Therefore, global variables are allowed to affect VN-Index, USD/VND returns, and exchange rate volatility contemporaneously, while Vietnam-specific variables are not assumed to contemporaneously affect global financial conditions. This recursive identification strategy is consistent with the view that Vietnam is a small open economy relative to global financial markets.

Impulse response functions are then used to trace the dynamic response of USD/VND returns and selected volatility to structural shocks over time. The SVAR results are used as supplementary evidence because they depend on the identifying ordering and weekly aggregation. Therefore, they are interpreted as dynamic shock-transmission patterns rather than definitive causal effects. The main volatility conclusions are drawn from the GARCH-family and EGARCH-X models, while the HAC and SVAR analyses provide robustness and transmission-channel evidence.

H. Summary of Hypothesis

Based on the research objective and the empirical modelling framework, this study develops a set of hypotheses on USD/VND exchange rate volatility and shock transmission. The hypotheses are designed to test whether USD/VND volatility is time-varying, whether GARCH-type models are appropriate, and whether global and domestic financial shocks help explain movements in exchange rate volatility.

These hypotheses are tested sequentially. The first three hypotheses are evaluated through preliminary diagnostics and GARCH-family model estimation. The fourth and fifth hypotheses are examined through the EGARCH-X specifications and HAC shock-volatility regressions. The sixth hypothesis is tested by comparing signed-shock and absolute-shock specifications. The seventh hypothesis is evaluated through the VN-Index coefficients in the EGARCH-X, HAC, and weekly SVAR frameworks.

TABLE III
RESEARCH HYPOTHESES

Hyp.	Statement	Main Test
H_1	USD/VND returns exhibit volatility clustering and conditional heteroskedasticity.	Ljung–Box, ARCH-LM
H_2	USD/VND exchange rate volatility is persistent over time.	GARCH-family models
H_3	USD/VND return innovations are fat-tailed and are better modelled using Student- t errors.	Jarque–Bera, Student- t parameter
H_4	Major exchange-rate policy events are associated with higher USD/VND volatility.	EGARCH-X, HAC
H_5	Global financial shocks affect USD/VND exchange rate volatility.	EGARCH-X, HAC, SVAR
H_6	Shock magnitudes explain USD/VND volatility more strongly than signed shock directions.	Signed vs. absolute shock models
H_7	Domestic financial-market shocks contribute to USD/VND exchange rate volatility.	VN-Index coefficients

IV. RESULTS

A. Preliminary Time-Series Diagnostic Results

This section reports the preliminary diagnostic results for the transformed model variables. The purpose of these tests is to evaluate whether the data exhibit the empirical properties required for volatility modeling, including stationarity, serial dependence, volatility clustering, conditional heteroskedasticity, non-normality, and absence of severe multicollinearity. These tests provide the statistical basis for using ARMA mean filtering, GARCH-family volatility models, EGARCH-X shock-transmission specifications, and supplementary HAC and SVAR analysis.

1) *Descriptive Statistics:* Table IV presents descriptive statistics for the main transformed variables used in the empirical analysis. The final modeling sample contains 4,006 daily observations for most variables over the 2010–2026 period, with 4,004 observations for oil returns due to missing values after transformation.

TABLE IV
DESCRIPTIVE STATISTICS FOR MODEL VARIABLES

Variable	Obs.	Mean	Std.	Skew.	Kurt.
fx_return	4006	0.0088	0.2129	6.4640	213.1828
dxy_return	4006	0.0057	0.4474	-0.0348	4.5347
vix_change	4006	-0.0026	1.8739	1.9111	32.2619
us10y_change	4006	0.0001	0.0535	0.0043	4.6246
vnindex_return	4006	0.0327	1.1763	-0.8105	7.2700
oil_return	4004	0.0113	2.7136	1.0326	43.5383

The descriptive statistics show that the average USD/VND return is close to zero, with a mean of 0.0088 percent. This is consistent with the behavior of daily exchange rate returns, where the average daily movement is small relative to occasional extreme changes. However, the distribution of USD/VND returns is highly non-normal. The series has strong positive skewness of 6.4640 and a very high Pearson kurtosis of 213.1828, indicating a heavy right tail and a high frequency of extreme observations. This pattern is consistent with a managed exchange rate environment, where daily movements are usually small but occasional discrete adjustments can generate large tail observations.

The standard deviation of `fx_return` is only 0.2129, which is substantially lower than that of `vnindex_return` and `oil_return`. This reflects the relatively managed nature of the USD/VND exchange rate compared with equity and commodity prices. Nevertheless, the high skewness and kurtosis of USD/VND returns show that extreme exchange rate adjustments still occur within the sample. The high kurtosis of VIX changes and oil returns also confirms that the external shock variables contain large tail events. These distributional features justify the use of fat-tailed innovations in the subsequent volatility models.

Table V reports the number of observations covered by the event-period dummy variables. The aggregate crisis dummy equals one for 599 daily observations, capturing major periods of exchange rate devaluation, the COVID-19 shock, and the 2022–2023 Federal Reserve tightening cycle.

TABLE V
EVENT DUMMY COUNTS

Dummy Variable	Number of Days
<code>vietnam_fx_devaluation_2010</code>	24
<code>vietnam_fx_devaluation_2011</code>	6
<code>covid_2020</code>	213
<code>fed_hiking_2022_2023</code>	356
<code>crisis_dummy</code>	599

2) *Stationarity Tests:* Table VI reports the ADF and KPSS stationarity test results. The ADF test rejects the null hypothesis of a unit root for all transformed variables at the 1 percent level. At the same time, the KPSS test does not reject the null hypothesis of stationarity for any variable, with p-values reported at the upper bound of 0.1000 for all series. The two tests therefore provide consistent evidence that the transformed variables are stationary.

TABLE VI
ADF AND KPSS STATIONARITY TEST RESULTS

Variable	ADF	ADF p	Lag	KPSS	KPSS p	Lag	Res.
<code>fx_return</code>	-32.4961	0.0000	3	0.1303	0.1000	5	I(0)
<code>dxy_return</code>	-62.4604	0.0000	0	0.0797	0.1000	11	I(0)
<code>vix_change</code>	-15.1685	0.0000	25	0.0109	0.1000	34	I(0)
<code>usl0y_change</code>	-47.1905	0.0000	1	0.2569	0.1000	8	I(0)
<code>vnindex_return</code>	-41.5835	0.0000	1	0.0724	0.1000	8	I(0)
<code>oil_return</code>	-11.9710	0.0000	23	0.0692	0.1000	1	I(0)

These results indicate that the variables used in the empirical models are integrated of order zero, I(0). Therefore, the return and change variables can be used directly in the ARMA, GARCH-family, EGARCH-X, HAC, and VAR/SVAR models without additional differencing. This is important because applying volatility models to non-stationary variables could produce misleading persistence and volatility estimates.

3) *Autocorrelation and Mean Dependence:* The ACF and PACF results indicate short-run dependence in USD/VND returns. Using the 95 percent confidence bound of approximately ± 0.0310 , significant ACF lags are found at lags 1, 3, 4, 5, 8, 9, 10, 12, 13, and 18. Significant PACF lags are found at lags 1, 2, 3, 4, 5, 8, 10, and 12. These patterns suggest that

USD/VND returns are not pure white noise in the conditional mean.

Table VII confirms this result formally. The Ljung–Box test rejects the null hypothesis of no autocorrelation at lags 5, 10, and 20, with all p-values equal to 0.0000. This provides direct support for including an ARMA-type conditional mean equation before estimating volatility models.

TABLE VII
LJUNG–BOX TEST ON USD/VND RETURNS

Lag	Q-stat.	p-value	Result
5	108.8015	0.0000	Autocorrelation detected
10	146.8841	0.0000	Autocorrelation detected
20	181.1506	0.0000	Autocorrelation detected

The presence of serial correlation means that the conditional mean cannot be ignored. Therefore, the subsequent modelling strategy first accounts for short-run mean dynamics before focusing on the conditional variance process.

4) *Volatility Clustering and ARCH Effects:* Table VIII reports Ljung–Box tests on absolute and squared USD/VND returns. The tests strongly reject the null hypothesis of no serial correlation for both volatility proxies at all reported lags. For absolute returns, the Ljung–Box statistic is 1007.3634 at lag 5, 1482.5224 at lag 10, and 1685.4318 at lag 20. For squared returns, the corresponding statistics are 48.8911, 83.5073, and 90.3285. All p-values are 0.0000.

TABLE VIII
LJUNG–BOX TESTS ON ABSOLUTE AND SQUARED RETURNS

Variable	Property	Lag	Q-stat.	Conclusion
<code>abs_return</code>	$ r_t $	5	1007.3634	Clustering detected
<code>abs_return</code>	$ r_t $	10	1482.5224	Clustering detected
<code>abs_return</code>	$ r_t $	20	1685.4318	Clustering detected
<code>squared_return</code>	r_t^2	5	48.8911	Clustering detected
<code>squared_return</code>	r_t^2	10	83.5073	Clustering detected
<code>squared_return</code>	r_t^2	20	90.3285	Clustering detected

Note: All reported p-values are 0.0000.

These results show that the magnitude of exchange rate movements is persistent over time. In economic terms, periods of large USD/VND movements tend to be followed by further large movements, while tranquil periods tend to persist. This pattern is the standard empirical motivation for using GARCH-family models rather than constant-variance models.

The ARCH-LM results in Table IX provide further evidence of conditional heteroskedasticity. The test rejects the null hypothesis of no ARCH effects for both constant-mean residuals and residuals from the compact OLS mean equation. For constant-mean residuals, the ARCH-LM statistic is 59.5223 with a p-value of 0.0000. For OLS mean equation residuals, the statistic is 61.3004 with a p-value of 0.0000.

TABLE IX
ARCH-LM TEST RESULTS

Residual Source	Lags	ARCH-LM Stat.	ARCH-LM p-value	F-stat.	Conclusion
Constant mean residuals	10	59.5223	0.0000	6.0256	ARCH effect detected
OLS mean equation residuals	10	61.3004	0.0000	6.2084	ARCH effect detected

Note: The F-test p-values are also 0.0000 for both residual sources.

The OLS mean equation used for this diagnostic check includes `dxy_return`, `vix_change`, `us10y_change`, `vnindex_return`, and `crisis_dummy`. The design matrix has full rank, with a condition number of 36.7073, indicating no singular matrix problem in this aggregate-dummy specification. However, the model has very low explanatory power, with $R^2 = 0.0052$ and adjusted $R^2 = 0.0040$. This confirms that the OLS mean equation is not a sufficient model for USD/VND volatility. Its role is only diagnostic: it shows that volatility dependence remains even after controlling for basic shock variables and crisis periods.

Table X reports Ljung–Box tests on squared OLS residuals. The null hypothesis of no serial correlation in squared residuals is rejected at lags 5, 10, and 20. This confirms that ARCH-type dependence remains after the mean equation.

TABLE X
LJUNG–BOX TESTS ON SQUARED OLS RESIDUALS

Lag	Q-stat.	p-value	Conclusion
5	51.0218	0.0000	ARCH/volatility clustering remains
10	86.9527	0.0000	ARCH/volatility clustering remains
20	94.2665	0.0000	ARCH/volatility clustering remains

Taken together, the Ljung–Box tests on volatility proxies, the ARCH-LM tests, and the squared-residual diagnostics provide strong evidence that USD/VND return volatility is time-varying. Therefore, GARCH-family volatility models are required.

5) *Normality Test*: Table XI reports the Jarque–Bera normality test for USD/VND returns. The null hypothesis of normality is strongly rejected, with a Jarque–Bera statistic of 7,383,227.0371 and a p-value of 0.0000. The return series has skewness of 6.4616 and Pearson kurtosis of 212.9190.

TABLE XI
JARQUE–BERA NORMALITY TEST FOR USD/VND RETURNS

Variable	JB Stat.	p-value	Skew.	Kurt.	Conclusion
<code>fx_return</code>	7383227.0371	0.0000	6.4616	212.9190	Non-normal

This result confirms that USD/VND returns are not normally distributed. The extremely high kurtosis indicates fat-tailed behaviour, meaning that large exchange rate movements occur more frequently than under a Gaussian distribution. Therefore, Student-*t* innovations are more appropriate than normal innovations in the GARCH-family models.

6) *Correlation and Multicollinearity Diagnostics*: Table XII reports the pairwise correlations among the main return and shock variables. The correlations among the explanatory

shock variables are generally moderate. The strongest correlations are between `vix_change` and `us10y_change` at -0.2336 , between `vix_change` and `oil_return` at -0.2176 , and between `us10y_change` and `oil_return` at 0.1705 . These values are not high enough to indicate severe pairwise multicollinearity.

TABLE XII
CORRELATION MATRIX OF MAIN MODEL VARIABLES

	<code>fx</code>	<code>dxy</code>	<code>vix</code>	<code>us10y</code>	<code>vni</code>	<code>oil</code>
<code>fx_return</code>	1.0000	0.0093	0.0026	0.0034	-0.0635	-0.0093
<code>dxy_return</code>	0.0093	1.0000	0.0854	0.1561	0.0013	-0.1096
<code>vix_change</code>	0.0026	0.0854	1.0000	-0.2336	-0.0757	-0.2176
<code>us10y_change</code>	0.0034	0.1561	-0.2336	1.0000	0.0423	0.1705
<code>vnindex_return</code>	-0.0635	0.0013	-0.0757	0.0423	1.0000	0.0445
<code>oil_return</code>	-0.0093	-0.1096	-0.2176	0.1705	0.0445	1.0000

The contemporaneous correlation between USD/VND returns and most shock variables is small. This does not mean that the shocks are irrelevant. Instead, it suggests that simple contemporaneous correlation is not sufficient to capture exchange rate risk transmission. This supports the use of conditional volatility models and dynamic VAR/SVAR analysis rather than relying only on static correlations.

Table XIII reports VIF values for the main explanatory variables. All VIF values are close to one, ranging from 1.0083 for `vnindex_return` to 1.1196 for `us10y_change`. These values are far below conventional warning thresholds.

TABLE XIII
VARIANCE INFLATION FACTOR RESULTS

Variable	VIF
<code>dxy_return</code>	1.0575
<code>vix_change</code>	1.1117
<code>us10y_change</code>	1.1196
<code>vnindex_return</code>	1.0083
<code>oil_return</code>	1.0839

The VIF results indicate that there is no severe multicollinearity among the explanatory shock variables. Therefore, the shock variables can be included jointly in the EGARCH-X and HAC specifications without serious collinearity concerns.

7) *Summary of Diagnostic Evidence*: The preliminary tests provide a consistent empirical foundation for the modelling strategy. First, the ADF and KPSS tests show that all transformed variables are stationary, allowing them to be used directly in the time-series models. Second, the ACF, PACF, and Ljung–Box tests show that USD/VND returns contain short-run serial dependence, which justifies the use of an ARMA conditional mean specification. Third, the Ljung–Box tests on absolute and squared returns, the ARCH-LM tests, and the squared OLS residual diagnostics confirm strong volatility clustering and conditional heteroskedasticity. Fourth, the Jarque–Bera test rejects normality and supports the use of Student-*t* innovations. Finally, the correlation and VIF results show no severe multicollinearity among the shock variables.

Overall, the diagnostic evidence supports the use of an autoregressive mean specification combined with GARCH-family volatility models under Student- t errors. It also supports extending the preferred volatility model into an EGARCH-X framework to examine whether global and domestic shocks help explain USD/VND exchange rate volatility.

B. Conditional Mean Model Selection

C. Conditional Mean Model Selection

Before estimating the conditional volatility models, the short-run dynamics in USD/VND returns are examined using alternative ARMA specifications. This step is necessary because GARCH-type models should be applied after filtering predictable movements in the conditional mean. If serial dependence remains in the return equation, the estimated volatility process may partly reflect misspecified mean dynamics rather than genuine conditional heteroskedasticity.

Table below reports the ARMA model selection results. The candidate models are compared using log-likelihood, AIC, BIC, HQIC, convergence status, and residual diagnostic tests. Among the estimated specifications, ARMA(3,2) provides the best overall performance. It achieves the lowest AIC of -1146.615 , the lowest BIC of -1102.546 , and the lowest HQIC of -1130.995 . Although several higher-order models, such as ARMA(3,5), ARMA(4,4), and ARMA(5,4), produce slightly higher log-likelihood values, they require more parameters and are penalized more heavily by the information criteria. Therefore, ARMA(3,2) offers the most efficient balance between model fit and parsimony.

The residual diagnostics also support the use of ARMA(3,2) as the conditional mean specification. The Ljung-Box p-values for the selected model are 0.324 at lag 10 and 0.136 at lag 20, indicating that the model removes most linear serial correlation in the USD/VND return series. The residual mean is close to zero, at 0.000013, and the residual standard deviation is 0.209425, suggesting that the fitted mean equation is not systematically biased. These results indicate that the remaining residuals are suitable for subsequent volatility modeling.

However, the diagnostic results also show that modeling the conditional mean alone is insufficient. The Ljung-Box tests on squared residuals remain significant at both lag 10 and lag 20, with p-values of 0.000001 and 0.000129, respectively. The ARCH-LM test is also highly significant, with a p-value of 0.000048. These findings imply that although ARMA(3,2) captures the short-run dependence in the mean equation, substantial conditional heteroskedasticity remains in the residuals. This is consistent with the presence of volatility clustering in USD/VND returns and provides direct motivation for estimating GARCH-family models in the next stage.

Based on these results, ARMA(3,2) is selected as the preferred conditional mean specification. The selected model is not treated as the final model of the study. Instead, it serves as a mean-filtering step before estimating the conditional variance process. The persistence of ARCH effects after ARMA filtering confirms that exchange rate volatility must be modeled explicitly using GARCH-type specifications.

TABLE XV
GARCH-FAMILY MODEL FIT COMPARISON

ID	Model	LogLik	AIC	BIC
M1	ARMA(3,2)-EGARCH(1,1)-t	3346.249	-6684.497	-6659.318
M2	ARMA(3,2)-EGARCH(1,1)-asym-t	3346.525	-6683.050	-6651.576
M3	ARMA(3,2)-GJR-GARCH(1,1)-t	3146.639	-6283.279	-6251.805
M4	ARMA(3,2)-GARCH(1,1)-t	3140.870	-6273.741	-6248.561

TABLE XVI
RESIDUAL DIAGNOSTICS OF GARCH-FAMILY MODELS

ID	LB Res.	LB Sq.	ARCH	Vol.	Full
M1	0.000	0.996	0.996	Yes	No
M2	0.000	0.996	0.996	Yes	No
M3	0.000	0.000	0.000	No	No
M4	0.000	0.000	0.000	No	No

Note: LB Res. and LB Sq. are Ljung-Box p-values for standardized residuals and squared standardized residuals. ARCH denotes the ARCH-LM p-value.

TABLE XIV
ARMA MEAN EQUATION SELECTION

Model	AIC	BIC	HQIC	LB(10)	LB(20)	ARCH-LM
ARMA(3,2)	-1146.615	-1102.546	-1130.995	0.324	0.136	0.000
ARMA(2,3)	-1144.639	-1100.571	-1129.019	0.266	0.101	0.000
ARMA(2,4)	-1145.384	-1095.019	-1127.532	0.366	0.159	0.000
ARMA(3,5)	-1146.029	-1083.073	-1123.714	0.830	0.353	0.000
ARMA(4,4)	-1146.014	-1083.059	-1123.700	0.839	0.365	0.000

The significant ARCH-LM statistic after ARMA filtering indicates that the remaining variation in USD/VND returns is dominated by conditional variance dynamics rather than by additional mean predictability. Therefore, the next stage estimates GARCH-family models using ARMA-filtered innovations.

D. GARCH-family Model Comparison

After selecting ARMA(3,2) as the conditional mean specification, the residuals from this model are used as the input series for the GARCH-family volatility models. This two-step structure separates the modeling of the conditional mean from the modeling of the conditional variance. Therefore, the GARCH-family models are estimated with a zero-mean specification because the short-run return dynamics have already been filtered out by the ARMA(3,2) model. The residual sample used for this stage contains 4,003 observations.

Table reports the comparison among four GARCH-family specifications: GARCH(1,1), GJR-GARCH(1,1), EGARCH(1,1), and asymmetric EGARCH(1,1), all estimated under the Student- t distribution. The Student- t distribution is used because the preliminary diagnostics indicate fat-tailed behavior in USD/VND exchange rate returns.

The results show that ARMA(3,2)-EGARCH(1,1) t is the preferred standard volatility model. It has the highest log-likelihood, at 3346.249, and the lowest AIC and BIC values, at -6684.497 and -6659.318 , respectively. Although the asymmetric EGARCH specification produces a slightly higher log-likelihood, it introduces an additional parameter and is penalized by both AIC and BIC. Therefore, the standard

EGARCH(1,1)- t specification provides a better balance between model fit and parsimony.

The diagnostic results further support the choice of EGARCH(1,1)- t . The model passes the volatility diagnostics, with Ljung-Box p-values on squared standardized residuals of 0.996 at lag 10 and 0.999 at lag 20. The ARCH-LM p-value is also 0.996, indicating that the model successfully removes the remaining ARCH effects from the ARMA residuals. This is an important result because the ARMA(3,2) model alone still left significant conditional heteroskedasticity in the residuals. After EGARCH estimation, the volatility dependence is largely absorbed by the conditional variance equation.

By contrast, the GARCH(1,1)- t and GJR-GARCH(1,1)- t models perform substantially worse. Their log-likelihood values are much lower, and their AIC and BIC values are far less favorable. More importantly, both models fail the volatility diagnostics: the Ljung-Box tests on squared standardized residuals and the ARCH-LM tests remain highly significant. This indicates that the symmetric GARCH and GJR-GARCH specifications do not fully capture the volatility clustering in USD/VND exchange rate returns.

The standardized residuals from the selected EGARCH model have a mean close to zero and a standard deviation close to one, suggesting that the model produces reasonably scaled residuals. However, the Ljung-Box tests on standardized residuals remain significant at both lag 10 and lag 20. This means that some linear dependence remains in the standardized residuals even after ARMA filtering and EGARCH volatility modelling. Therefore, the selected EGARCH model should not be interpreted as fully eliminating all serial dependence in the data. Its strength is that it captures the conditional volatility process effectively, which is the main objective of this stage of the analysis.

The estimated EGARCH parameters provide further evidence of persistent volatility. The magnitude parameter α_1 is positive and statistically significant, with a coefficient of 0.533 and a p-value below 0.01. This indicates that large standardized shocks have a strong effect on current volatility. The persistence parameter β_1 is also positive and highly significant, with a coefficient of 0.960. This value implies that volatility shocks decay slowly over time, confirming strong volatility persistence in the USD/VND exchange rate. The estimated degrees-of-freedom parameter of the Student- t distribution is 2.753, which confirms very heavy-tailed innovations and supports the use of a fat-tailed error distribution rather than a normal distribution.

Overall, the GARCH-family comparison shows that EGARCH(1,1)- t is the most suitable baseline volatility specification for USD/VND returns. It provides the best information criteria among the parsimonious models and successfully removes the remaining ARCH effects from the ARMA-filtered residuals. This model is therefore used as the baseline conditional volatility framework before extending the analysis to the EGARCH-X shock transmission models.

E. Selected Conditional Volatility Series

Based on the GARCH-family comparison, the ARMA(3,2)-EGARCH(1,1)- t model is selected as the baseline volatility model. The fitted conditional variance and conditional volatility from this model are then extracted for further analysis. The selected volatility source is therefore fixed as ARMA(3,2)-EGARCH(1,1)- t , ensuring that the subsequent shock-volatility regressions and weekly SVAR analysis are based on the best-performing standard volatility specification.

The extracted conditional volatility series is denoted by $\hat{\sigma}_t^{sel}$, while the corresponding conditional variance is denoted by $\hat{h}_t^{sel}$. The logarithmic transformations are also constructed as follows:

$$\begin{aligned} selected_cond_vol_t &= \hat{\sigma}_t^{sel}, \\ selected_cond_var_t &= \hat{h}_t^{sel}, \\ log_selected_cond_vol_t &= \ln(\hat{\sigma}_t^{sel}), \\ log_selected_cond_var_t &= \ln(\hat{h}_t^{sel}). \end{aligned} \quad (22)$$

The logarithm of selected conditional volatility is used in the supplementary HAC and SVAR analyses because it provides a more stable scale and reduces the influence of extremely large volatility values. This transformation is particularly useful when the estimated volatility series is used as a dependent variable in regression-based analysis or as an endogenous variable in a VAR/SVAR system.

The extracted volatility series serves two purposes in the empirical analysis. First, it provides a model-based measure of USD/VND exchange rate risk. Second, it is used as the volatility target in the supplementary HAC regression and weekly SVAR framework, allowing the study to examine whether global and domestic shocks are associated with both exchange rate returns and estimated exchange rate volatility.

F. EGARCH-X Shock Transmission Results

After selecting ARMA(3,2)-EGARCH(1,1)- t as the preferred standard volatility model, the analysis is extended to EGARCH-X specifications in order to examine whether major policy events and global financial shocks help explain USD/VND exchange rate volatility. This step is central to the research question because the standard EGARCH model captures volatility persistence and fat-tailed innovations, but it does not directly identify which external or domestic variables are associated with changes in conditional volatility.

The EGARCH-X models are estimated using the ARMA(3,2) residuals as the filtered return innovation series. Six specifications are considered. The first is the baseline EGARCH model without external regressors. The second includes only the February 2010 Vietnam devaluation dummy. The remaining specifications include lagged signed shocks or lagged absolute shocks, with and without the devaluation dummy. The shock variables include changes in the VIX, changes in the U.S. 10-year Treasury yield, DXY returns, oil returns, and VN-Index returns. Lagged shocks are used because the USD/VND market may not fully absorb global and domestic financial

shocks contemporaneously, especially in a managed exchange rate environment.

Table XVII reports the EGARCH-X model comparison. All estimated EGARCH-X specifications converge successfully and pass the core volatility diagnostics. In particular, the Ljung–Box tests on squared standardized residuals and the ARCH-LM tests are insignificant across all EGARCH-X models, indicating that the conditional variance equations successfully absorb the remaining volatility clustering. However, the Ljung–Box tests on standardized residuals remain significant. Therefore, the models should be interpreted as successful volatility models, but not as models that fully eliminate all residual serial dependence.

Based on the BIC criterion, the preferred EGARCH-X specification is the model including only the February 2010 devaluation dummy. This model improves on the no-X EGARCH benchmark. Its log-likelihood increases from 3342.353 to 3350.110, while AIC improves from -6674.707 to -6688.221 and BIC improves from -6643.240 to -6650.461 . This result indicates that the February 2010 policy devaluation episode contains important information for explaining USD/VND volatility. The selection of this parsimonious specification by BIC also suggests that a small number of major exchange-rate policy events can explain part of the volatility dynamics without requiring a large number of additional shock variables.

The richer shock-augmented specifications provide additional evidence on global shock transmission. The model with lagged absolute shocks and the devaluation dummy achieves the highest log-likelihood, at 3360.431, and the lowest AIC, at -6696.862 . However, because it includes more parameters, its BIC is less favorable than the parsimonious devaluation-dummy specification. This distinction is important. The BIC result favours the simpler model, while the AIC and coefficient evidence from the richer model suggest that lagged shock magnitudes contain useful information about USD/VND volatility.

TABLE XVII
EGARCH-X MODEL COMPARISON

Model	Par.	LogLik	AIC	BIC
No X	5	3342.353	-6674.707	-6643.240
Devaluation dummy only	6	3350.110	-6688.221	-6650.461
Absolute shocks	11	3352.934	-6683.868	-6614.644
Absolute shocks + devaluation	12	3360.431	-6696.862	-6621.346
Signed shocks	11	3348.887	-6675.775	-6606.551
Signed shocks + devaluation	12	3355.357	-6686.715	-6611.198

Note: Par. denotes the number of estimated parameters. Bold values indicate the best model according to each criterion. All models pass the core volatility diagnostics based on Ljung–Box tests on squared standardized residuals and ARCH-LM tests.

Table XVIII reports the key coefficients from the EGARCH-X specifications. The selected devaluation-dummy model confirms the main properties already observed in the standard EGARCH model. The absolute shock coefficient is positive and statistically significant, with a coefficient of 0.531 and a p-value below 0.01. This indicates that large innovations in the USD/VND return process increase conditional volatility. The

volatility persistence coefficient is also highly significant, with a coefficient of 0.958, showing that exchange rate volatility remains persistent even after controlling for the devaluation dummy. The Student-*t* degrees-of-freedom parameter is 2.781 and statistically significant, confirming that USD/VND innovations remain strongly fat-tailed.

The asymmetry coefficient is positive but statistically insignificant in the selected EGARCH-X model. This suggests that the main volatility pattern in the USD/VND market is driven more by the size of shocks and volatility persistence than by a systematic difference between positive and negative shocks. This finding is reasonable for a managed exchange rate context, where large exchange rate adjustments and policy episodes may matter more than the sign of small daily innovations.

The February 2010 devaluation dummy enters the selected model at the optimizer boundary. Therefore, its exact coefficient magnitude and statistical significance should not be over-interpreted. The correct interpretation is that the dummy identifies an exceptional policy-related volatility episode that improves the model's fit and information criteria. It should be treated as an event-control result rather than as a precisely estimated marginal effect.

The broader shock specifications provide more direct evidence on shock transmission. In the lagged absolute-shock model, the coefficient on absolute oil-price shocks is positive and statistically significant at the 5% level. This suggests that larger oil-market movements are associated with higher USD/VND exchange rate volatility. In the absolute-shock model with the devaluation dummy, the DXY and oil coefficients are both positive and significant at the 10% level. These results indicate that the magnitude of global dollar and oil-market shocks contains useful information for explaining exchange rate volatility, although the evidence is weaker than the evidence for volatility persistence and the devaluation episode.

By contrast, the signed-shock specifications provide limited evidence that the direction of global shocks systematically affects USD/VND volatility. Most signed shock coefficients are statistically insignificant. This implies that the size of external shocks is more informative than their sign. In other words, large global market movements appear more relevant for exchange rate volatility than whether those movements are positive or negative on a given day.

TABLE XVIII
KEY EGARCH-X COEFFICIENT ESTIMATES

Spec.	Parameter	Coef.	SE	t-stat.	p-val.
Selected	ω	-0.164	0.027	-6.020	0.000
Selected	α	0.531	0.044	12.117	0.000
Selected	γ	0.017	0.021	0.821	0.412
Selected	β	0.958	0.006	149.810	0.000
Selected	ν	2.781	0.146	19.076	0.000
Abs. shocks	$ Z_{t-1}^{OIL} $	0.036	0.017	2.149	0.032
Abs. + dev.	$ Z_{t-1}^{DXY} $	0.052	0.029	1.792	0.073
Abs. + dev.	$ Z_{t-1}^{OIL} $	0.035	0.018	1.928	0.054
Abs. + dev.	Crisis dummy	-0.084	0.024	-3.522	0.000

The crisis dummy has a negative and statistically significant

coefficient in the richer specifications. This result should be interpreted cautiously. It does not imply that crisis periods reduce exchange rate risk in a general economic sense. Rather, after controlling for the EGARCH volatility process, lagged shock magnitudes, and the devaluation event, the crisis dummy captures the remaining average shift in log variance during the defined crisis windows. In a managed exchange rate regime, policy intervention and exchange rate smoothing may cause crisis-period volatility to appear lower after other shocks are controlled for. Therefore, this coefficient is best interpreted as a control effect rather than as the main economic finding.

Overall, the EGARCH-X results support three main conclusions. First, USD/VND volatility is highly persistent and fat-tailed, as shown by the significant α , β , and Student- t parameters. Second, the February 2010 devaluation episode is an important volatility event and improves the EGARCH specification relative to the no- X benchmark. Third, the richer shock specifications provide evidence that the magnitude of selected global shocks, especially oil-price shocks and to a weaker extent DXY shocks, is associated with higher exchange rate volatility. These findings support the shock-transmission interpretation of the paper, while also showing that the strongest and most robust evidence remains concentrated in volatility persistence, fat-tailed innovations, and major exchange-rate policy episodes.

G. HAC Shock-Volatility Regression Results

After estimating the EGARCH-X specifications, a supplementary shock-volatility regression is conducted using heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The dependent variable is the logarithm of the selected conditional volatility series extracted from the ARMA(3,2)-EGARCH(1,1)- t model. This step is not intended to replace the EGARCH-X framework. Instead, it provides an additional reduced-form check on whether lagged global and domestic shocks are associated with the model-implied volatility of the USD/VND exchange rate.

Two HAC specifications are estimated. The first model uses signed lagged shocks, while the second model uses absolute lagged shocks. The signed-shock model tests whether the direction of shocks matters for exchange rate volatility. The absolute-shock model tests whether the magnitude of shocks matters regardless of direction. This distinction is important because exchange rate volatility may respond more strongly to the size of external disturbances than to whether those disturbances are positive or negative.

Table XIX reports the model summary. The absolute-shock specification performs better than the signed-shock specification. The adjusted R^2 increases from 0.0036 in the signed-shock model to 0.0333 in the absolute-shock model. The AIC and BIC values are also lower for the absolute-shock model. Although the explanatory power remains modest, this is expected because daily exchange rate volatility is highly persistent and partly driven by unobserved market and policy factors. The improvement in the absolute-shock model indicates

that volatility is more closely related to the magnitude of shocks than to their signed direction.

TABLE XIX
HAC SHOCK-VOLATILITY REGRESSION SUMMARY

Model	Obs.	R^2	Adj. R^2	AIC	BIC
HAC signed shocks	3997	0.0054	0.0036	8118.932	8169.278
HAC absolute shocks	3997	0.0350	0.0333	7998.136	8048.483

Table XX presents the coefficient estimates. In the signed-shock specification, none of the global or domestic shock variables is statistically significant. The coefficients on lagged VIX changes, U.S. 10-year yield changes, DXY returns, oil returns, and VN-Index returns are all insignificant. This suggests that the direction of these shocks does not provide strong explanatory power for USD/VND volatility once robust standard errors are applied. The only strongly significant variable in the signed-shock model is the February 2010 devaluation dummy, which has a positive coefficient of 1.248 and is significant at the 1% level.

The absolute-shock specification produces much stronger results. Lagged absolute changes in the U.S. 10-year Treasury yield, DXY returns, oil returns, and VN-Index returns are all positive and statistically significant at the 1% level. The coefficient on $|Z_{t-1}^{US10Y}|$ is 0.086, indicating that larger U.S. interest-rate movements are associated with higher USD/VND volatility. The coefficient on $|Z_{t-1}^{DXY}|$ is 0.083, showing that larger U.S. dollar movements are also linked to higher exchange rate volatility. The coefficient on $|Z_{t-1}^{OIL}|$ is 0.064, suggesting that oil-market disturbances contribute to volatility in the USD/VND market. The coefficient on $|Z_{t-1}^{VN}|$ is 0.073, indicating that domestic stock-market volatility is also associated with higher exchange rate volatility.

The coefficient on lagged absolute VIX changes is negative but statistically insignificant. Therefore, the HAC regression does not provide strong evidence that VIX shocks directly explain USD/VND volatility in this specification. This does not mean that global risk aversion is irrelevant in general; rather, within this reduced-form daily regression, VIX changes do not have a robust independent effect after controlling for U.S. yields, DXY, oil prices, VN-Index movements, and event dummies.

The February 2010 devaluation dummy is positive and highly significant in both HAC specifications. Its coefficient is 1.248 in the signed-shock model and 1.231 in the absolute-shock model, both significant at the 1% level. This result is consistent with the EGARCH-X evidence that the 2010 devaluation event represents an important exchange-rate volatility episode. Unlike the EGARCH-X coefficient on the devaluation dummy, which reaches the optimizer boundary and therefore should not be interpreted precisely, the HAC regression provides a clearer reduced-form indication that the devaluation event is strongly associated with a higher level of estimated conditional volatility.

The crisis dummy is negative in both specifications and becomes weakly significant in the absolute-shock model at the

TABLE XX
HAC SHOCK-VOLATILITY REGRESSION COEFFICIENTS

Model	Variable	Coef.	HAC Std. Error	t-stat.	p-value	Sig.
Signed shocks	Constant	-1.990	0.025	-81.004	0.000	***
Signed shocks	Z_{t-1}^{VIX}	-0.006	0.010	-0.621	0.535	
Signed shocks	Z_{t-1}^{US10Y}	-0.012	0.011	-1.095	0.274	
Signed shocks	Z_{t-1}^{DXY}	-0.016	0.010	-1.548	0.122	
Signed shocks	Z_{t-1}^{OIL}	-0.004	0.009	-0.405	0.686	
Signed shocks	Z_{t-1}^{VNI}	-0.021	0.013	-1.591	0.112	
Signed shocks	Crisis dummy	-0.096	0.087	-1.102	0.271	
Signed shocks	Devaluation dummy	1.248	0.086	14.429	0.000	***
Absolute shocks	Constant	-2.183	0.037	-59.539	0.000	***
Absolute shocks	$ Z_{t-1}^{VIX} $	-0.024	0.021	-1.157	0.247	
Absolute shocks	$ Z_{t-1}^{US10Y} $	0.086	0.022	3.990	0.000	***
Absolute shocks	$ Z_{t-1}^{DXY} $	0.083	0.017	4.956	0.000	***
Absolute shocks	$ Z_{t-1}^{OIL} $	0.064	0.015	4.139	0.000	***
Absolute shocks	$ Z_{t-1}^{VNI} $	0.073	0.021	3.410	0.001	***
Absolute shocks	Crisis dummy	-0.169	0.089	-1.892	0.058	
Absolute shocks	Devaluation dummy	1.231	0.089	13.855	0.000	***

Notes: The dependent variable is $\log(\hat{\sigma}_t^{sel})$, where $\hat{\sigma}_t^{sel}$ is the selected conditional volatility from the ARMA(3,2)-EGARCH(1,1)- t model. HAC standard errors are computed using five lags. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

10% level. This coefficient should be interpreted cautiously. It does not imply that crisis periods reduce exchange rate risk in an economic sense. Rather, after controlling for lagged shock magnitudes and the specific devaluation event, the crisis dummy captures the remaining average difference in estimated volatility during the defined crisis windows. In a managed exchange rate environment, policy smoothing and administrative intervention may reduce observed daily exchange rate variability during some crisis periods, even when underlying macro-financial pressure is elevated.

Overall, the HAC results reinforce the main interpretation from the EGARCH-X analysis. The evidence is stronger for absolute shock magnitudes than for signed shocks. U.S. interest-rate shocks, U.S. dollar movements, oil-price shocks, and domestic stock-market shocks are positively associated with USD/VND conditional volatility when measured in absolute terms. The devaluation dummy remains the most robust event-related variable. Therefore, the HAC regression supports the view that USD/VND volatility is linked not only to its own persistence, but also to the magnitude of global and domestic financial disturbances

H. Weekly SVAR and Dynamic Shock Transmission

The final stage of the empirical analysis uses weekly VAR/SVAR models to examine the dynamic transmission of global and domestic shocks to the USD/VND exchange rate and its selected conditional volatility. This analysis is supplementary to the EGARCH-X and HAC results. While the EGARCH-X model directly estimates shock effects in the conditional variance equation, the SVAR framework provides additional evidence on whether shocks propagate dynamically through the system over time.

Two weekly SVAR systems are estimated. The first model examines shock transmission to USD/VND returns, while the second examines shock transmission to the logarithm of selected conditional volatility. The daily data are aggregated to

weekly frequency to reduce high-frequency noise and improve the stability of the VAR system. The exchange rate return model includes 831 weekly observations, six variables, and four lags. The selected volatility model also includes 831 weekly observations and six variables, but uses one lag. Both systems are successfully estimated, indicating that the weekly SVAR framework is computationally stable for the sample.

Before interpreting the structural impulse responses, Granger causality tests are used to examine whether the lagged shock variables contain predictive information for USD/VND returns or selected exchange rate volatility. These tests should be interpreted as predictive evidence, not as proof of structural causality.

Table XXI reports the Granger causality results. For the USD/VND return system, the U.S. 10-year Treasury yield is the only statistically significant predictor at the 5% level, with a p-value of 0.000015. This result indicates that changes in U.S. interest-rate conditions contain useful predictive information for weekly USD/VND returns. The remaining variables, including VIX, DXY, oil prices, and VN-Index returns, are not statistically significant predictors of weekly exchange rate returns in this system.

For the selected volatility system, oil-price shocks are statistically significant at the 5% level, with a p-value of 0.026544. This means that lagged oil-price movements help predict the selected conditional volatility of USD/VND. In contrast, VIX, U.S. Treasury yields, DXY, and VN-Index returns do not show significant predictive power for selected volatility in the weekly SVAR system. This finding is consistent with the earlier EGARCH-X and HAC evidence that oil-market shocks are among the more relevant global shock channels for USD/VND volatility.

TABLE XXI
WEEKLY SVAR GRANGER CAUSALITY TESTS

Target	Shock	Lag	p-value	Sig. 5%
fx_return	vix_change	4	0.1395	No
fx_return	$us10y_change$	4	0.0000	Yes
fx_return	dxy_return	4	0.0980	No
fx_return	oil_return	4	0.3165	No
fx_return	$vnindex_return$	4	0.1860	No
$log_selected_cond_vol$	vix_change	1	0.8444	No
$log_selected_cond_vol$	$us10y_change$	1	0.2455	No
$log_selected_cond_vol$	dxy_return	1	0.5250	No
$log_selected_cond_vol$	oil_return	1	0.0265	Yes
$log_selected_cond_vol$	$vnindex_return$	1	0.6393	No

Notes: Granger causality tests evaluate whether lagged shock variables help predict the target variable within the weekly VAR system. Statistical significance indicates predictive content, not structural causality.

The structural impulse response results provide additional evidence on the dynamic response of USD/VND returns to identified shocks. The available impulse response estimates show that a VIX shock produces a positive response in USD/VND returns at impact and remains positive over the following four weeks. The response is 0.0859 at horizon 0, then declines to 0.0438 after one week, before fluctuating between 0.0390 and 0.0568 over horizons two to four. This pattern

suggests that global risk-aversion shocks may generate a short-run depreciation pressure on the Vietnamese dong, but the response is moderate and does not display a sharply increasing dynamic effect.

TABLE XXII
STRUCTURAL IMPULSE RESPONSE OF USD/VND RETURN TO VIX SHOCK

Response	Horizon	Structural IRF
<i>fx_return to vix_change</i>	0	0.0859
<i>fx_return to vix_change</i>	1	0.0438
<i>fx_return to vix_change</i>	2	0.0568
<i>fx_return to vix_change</i>	3	0.0390
<i>fx_return to vix_change</i>	4	0.0481

Notes: Horizon is measured in weeks. The table reports the structural response of USD/VND returns to a VIX shock from the weekly SVAR system.

Overall, the weekly SVAR results provide selective but useful support for the shock-transmission interpretation of the study. The Granger causality tests show that U.S. interest-rate shocks are relevant for predicting USD/VND returns, while oil-price shocks are relevant for predicting selected exchange rate volatility. The impulse response evidence also suggests that global risk-aversion shocks can affect USD/VND returns in the short run. However, most shock variables are not statistically significant in the weekly Granger tests. Therefore, the SVAR evidence should be interpreted as supplementary rather than as the main basis for the paper's conclusions. The strongest empirical evidence remains in the GARCH-family, EGARCH-X, and HAC results, while the weekly SVAR analysis adds dynamic evidence on how selected global shocks are transmitted to exchange rate returns and volatility.

V. DISCUSSIONS AND CONCLUSIONS

This study examines the transmission of global and domestic financial shocks to USD/VND exchange rate volatility using daily data from 2010 to 2026. The empirical strategy combines ARMA mean filtering, GARCH-family volatility modelling, EGARCH-X shock transmission analysis, HAC shock-volatility regressions, and weekly SVAR evidence. The results provide a consistent picture: USD/VND exchange rate volatility is persistent, fat-tailed, strongly affected by large exchange rate innovations, and partly linked to global and domestic financial disturbances.

The first important result is that the conditional mean of USD/VND returns is best represented by an ARMA(3,2) specification. This model provides the lowest AIC and BIC among the clean residual specifications and removes most linear serial dependence in the return equation. However, the residual diagnostics show that ARCH effects remain highly significant after ARMA filtering. This means that modelling the mean equation alone is insufficient. The remaining variation in USD/VND returns is dominated by time-varying conditional volatility, which justifies the use of GARCH-family models.

The GARCH-family comparison shows that ARMA(3,2)-EGARCH(1,1)- t is the most suitable baseline volatility model. It has the strongest information criteria among the standard GARCH specifications and successfully removes remaining

ARCH effects from the ARMA-filtered residuals. The estimated EGARCH parameters show strong volatility persistence, while the Student- t degrees-of-freedom parameter confirms that USD/VND innovations are fat-tailed. These results support the view that USD/VND exchange rate risk is not constant over time and that extreme movements occur more frequently than under a normal distribution.

The EGARCH-X results show that the February 2010 devaluation episode is an important volatility event. The BIC-preferred EGARCH-X specification includes the devaluation dummy and improves on the no- X EGARCH benchmark. This suggests that major exchange-rate policy events contain important information for explaining USD/VND volatility. However, the devaluation coefficient reaches an optimizer boundary, so its exact magnitude should not be interpreted as a precise marginal effect. The appropriate interpretation is that the devaluation episode represents an exceptional volatility event rather than a smoothly estimated continuous shock effect.

The richer EGARCH-X specifications provide additional but more selective evidence of global shock transmission. The models with lagged absolute shocks indicate that the magnitude of oil-price shocks, and to a weaker extent DXY shocks, is associated with higher USD/VND volatility. In contrast, most signed shock variables are statistically insignificant. This implies that the size of external disturbances matters more than their direction. For the USD/VND market, large global market movements appear more relevant for volatility than whether the daily shock is positive or negative.

The HAC regression results strengthen this interpretation. The absolute-shock HAC model performs better than the signed-shock model, with a higher adjusted R^2 and lower AIC/BIC. Lagged absolute movements in the U.S. 10-year Treasury yield, DXY, oil prices, and VN-Index are all positive and statistically significant. This provides reduced-form evidence that both global and domestic shock magnitudes are associated with higher estimated USD/VND conditional volatility. The devaluation dummy remains strongly positive and significant in the HAC regression, confirming its role as a major exchange-rate volatility event.

The weekly SVAR results provide supplementary dynamic evidence. The Granger causality tests show that U.S. 10-year Treasury yield changes help predict weekly USD/VND returns, while oil-price shocks help predict selected conditional volatility. These findings are broadly consistent with the HAC and EGARCH-X results, especially regarding the relevance of global interest-rate and oil-market shocks. However, most other variables are not statistically significant in the weekly Granger tests. Therefore, the SVAR results should be interpreted as supporting evidence rather than as the main empirical basis of the paper.

Overall, the findings suggest that USD/VND exchange rate volatility is shaped by three main forces. First, volatility is highly persistent and strongly driven by past exchange rate shocks. Second, major exchange-rate policy events, especially the February 2010 devaluation, are important volatility episodes. Third, global and domestic shock magnitudes, particularly

U.S. interest-rate movements, DXY movements, oil-price shocks, and VN-Index movements, are associated with higher exchange rate volatility in the supplementary HAC evidence. The results therefore support a shock-transmission interpretation of USD/VND volatility, but the evidence is stronger for volatility persistence, fat tails, and major event effects than for a broad causal effect of every individual global shock.

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